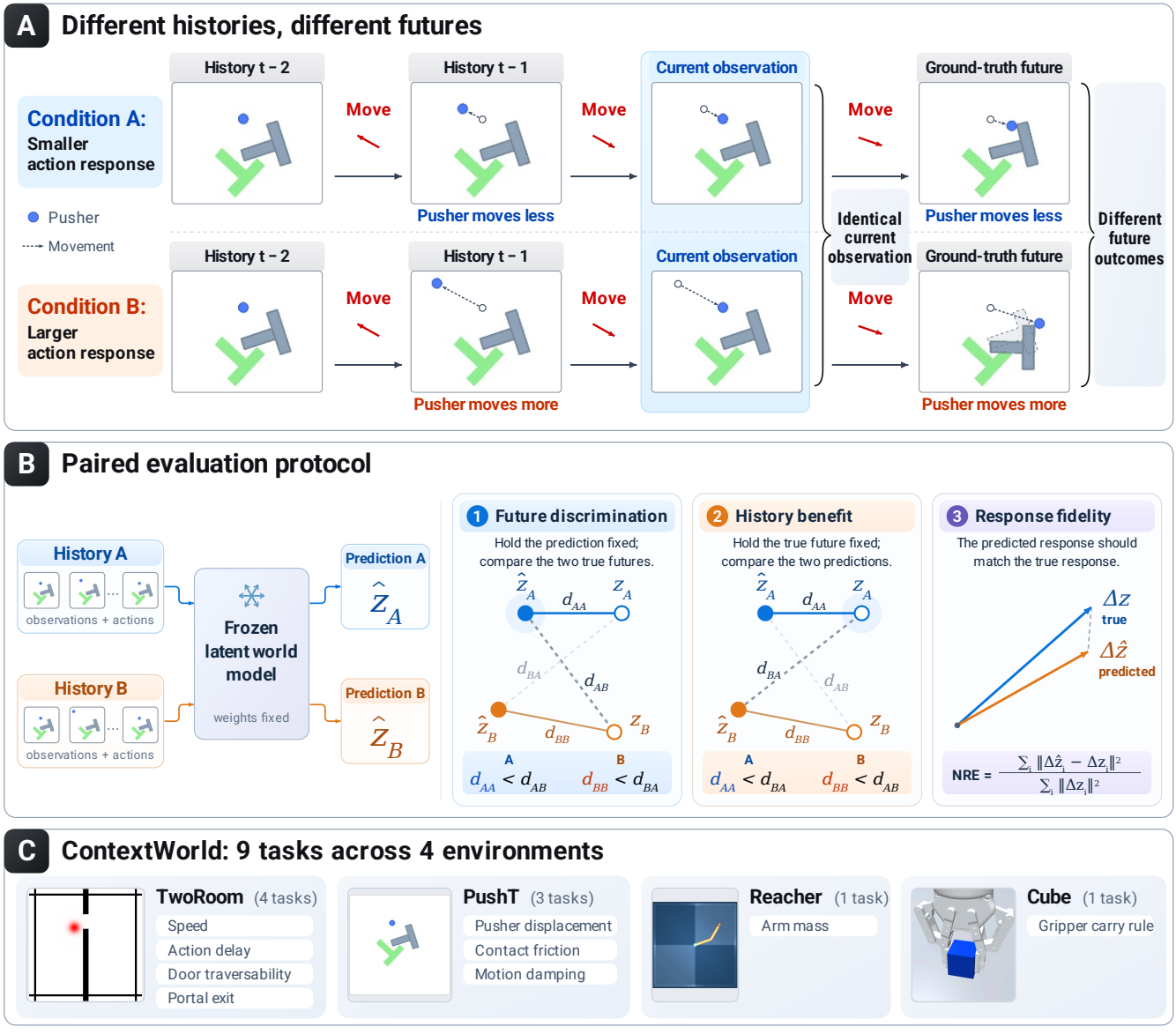

A
Different histories, different futures
Condition A:
Smaller action response
Pusher
Movement
Condition B:
Larger action response
History t − 2
History t − 1
Current observation
Ground-truth future
Move
Pusher moves less
Pusher moves more
Identical current observation
Different future outcomes
B
Paired evaluation protocol
History A
observations + actions
History B
Frozen latent world model
weights fixed
Prediction A
Prediction B
1 Future discrimination
Hold the prediction fixed; compare the two true futures.
2 History benefit
Hold the true future fixed; compare the two predictions.
3 Response fidelity
The predicted response should match the true response.
true
predicted
NRE =
C
ContextWorld: 9 tasks across 4 environments
TwoRoom (4 tasks)
Speed
Action delay
Door traversability
Portal exit
PushT (3 tasks)
Pusher displacement
Contact friction
Motion damping
Reacher (1 task)
Arm mass
Cube (1 task)
Gripper carry rule